

ARITHMETIC RIGIDITY AND GROWTH IN INTEGER-SIDED CYCLIC POLYGONS

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ABSTRACT. For a positive integer n , let $a(n)$ be the largest number of sides of a simple cyclic polygon of circumradius n whose side lengths are positive integers. We study two markedly different regimes in the arithmetic of the radius. At semiprime radii the classification is exact: $a(pq)$ is always 6, 8, or 18, with the exceptional values governed by Gaussian and Eisenstein splitting. By contrast, fixed prime support does not force boundedness. For every integer $N \geq 2$,

$$a(N^e) \longrightarrow \infty, \quad \limsup_{e \rightarrow \infty} \frac{a(N^e)}{e} \geq 2.$$

An explicit construction based on a single quadratic angle tower and odd–odd Farey neighbors gives, for example, $a(8192) \geq 38$. We also prove a parity theorem: if n has at most one distinct prime divisor $q \equiv 1 \pmod{4}$, then $a(n)$ is even. In particular, every prime-power maximum $a(p^e)$ is even. The proofs combine the basis and splitting theorem for pure geodetic angles of Conway, Radin, and Sadun with ideal-class arguments, bounded exact certification in the semiprime case, and a one-dimensional Farey optimization in the Gaussian class.

1. INTRODUCTION

A cyclic polygon is a polygon whose vertices lie on a common circle. We fix the circumradius to be a positive integer and require every side length to be a positive integer. Classical arithmetic work on cyclic polygons has largely concerned area formulas and algebraic relations among prescribed side lengths; see Robbins [6] and Miller Maley, Robbins, and Roskies [3]. Here the radius itself is fixed and integral, and the quantity to be maximized is the number of sides.

For $n \geq 1$ and $1 \leq s \leq 2n$, put

$$(1) \quad \alpha_n(s) = \arcsin\left(\frac{s}{2n}\right) \in (0, \pi/2].$$

Thus $2\alpha_n(s)$ is the minor central angle subtended by a chord of length s .

Definition 1.1. Let $a(n)$ be the largest integer k for which there is a simple cyclic k -gon of circumradius n with positive integer side lengths.

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The function $a(n)$ is OEIS A398490 [5]; the corresponding counting triangle $T(n, k)$ is OEIS A398489 [4]. The regular hexagon gives $a(n) \geq 6$ for every n . Already among radii with only two prime factors the values vary considerably:

$$a(49) = 18, \quad a(65) = 8, \quad a(91) = 18, \quad a(361) = 6.$$

The jumps are arithmetic rather than metric. For a prime $r \equiv 1 \pmod{3}$, choose the solution

$$(2) \quad s_r^2 + 3u_r^2 = 4r^2$$

with $s_r > 0$ minimal, and put

$$(3) \quad x_r = \arcsin\left(\frac{s_r}{2r}\right) \in (0, \pi/6).$$

Theorem 1.2 (Semiprime rigidity). *Let $p \leq q$ be primes. Then*

$$a(pq) = \begin{cases} 18, & p, q \text{ are odd, } p \equiv q \equiv 1 \pmod{3}, \\ & x_p + x_q < \pi/6, \\ 8, & p < q, \quad p \equiv q \equiv 1 \pmod{4}, \quad \text{and the first case fails,} \\ 6, & \text{otherwise.} \end{cases}$$

Moreover, $x_p + x_q < \pi/6$ is equivalent to

$$(4) \quad 3(u_p s_q + u_q s_p) < 3u_p u_q - s_p s_q.$$

In particular,

$$a(pq) \in \{6, 8, 18\}, \quad a(2p) = a(3p) = 6$$

for every prime p .

The semiprime classification gives a rigid picture when the radius has only two prime factors. Fixed prime support, however, does not force boundedness: increasing exponents alone can produce arbitrarily many sides.

Theorem 1.3 (Growth along powers). *For every integer $N \geq 2$,*

$$a(N^e) \longrightarrow \infty \quad (e \rightarrow \infty),$$

and more quantitatively

$$(5) \quad \limsup_{e \rightarrow \infty} \frac{a(N^e)}{e} \geq 2.$$

The proof of Theorem 1.3 starts from a suitable quadratic angle. Powers of a single quadratic integer produce integer chords at the radii N^e . A Stern–Brocot path then supplies infinitely many pairs of folded angles whose irrational parts cancel exactly. For $N = 2$, one such pair gives the concrete thirty-eight-gon

$$\{896^{[26]}, 2339^{[12]}\}$$

of radius $8192 = 2^{13}$.

Nevertheless, a parity restriction remains. Let

$$\omega_1(n) = \#\{q : q \mid n, \quad q \equiv 1 \pmod{4}\},$$

where distinct primes are counted once.

Theorem 1.4 (Parity in Gaussian rank one). *If $\omega_1(n) \leq 1$, then $a(n)$ is even. In particular,*

$$a(p^e) \equiv 0 \pmod{2}$$

for every prime p and every $e \geq 1$.

The three theorems arise from the same arithmetic decomposition mechanism. The Conway–Radin–Sadun splitting separates a polygon relation into squarefree arithmetic classes. With two prime factors, the resulting component geometry is finite and leads to the complete values 6, 8, 18. Along N^e , repeated powers of one quadratic angle create an expanding supply of integer chords, so the maximum becomes unbounded. In the Gaussian class with only one split rational prime, however, the optimization remains one-dimensional and is solved exactly by Farey neighbors; this forces the maximum to be even.

Section 2 gives the geometric criterion and scaling. Section 3 introduces geodetic splitting and derives the non-Gaussian parity principle and the odd-sided criterion. Section 4 proves the semiprime classification. Section 5 proves growth along every fixed power sequence, and Section 6 proves the parity theorem. Computer assistance is confined to bounded finite checks in the semiprime argument: the odd structural certificate and the even ideal-class verification. These checks are described in Section 7; the proofs of the growth and parity theorems do not use computation.

2. THE GEOMETRIC CRITERION

Lemma 2.1 (Minor- and major-arc criterion). *A multiset of positive integers $\{s_1, \dots, s_k\}$, with $s_i \leq 2n$, is realizable as the side-length multiset of a simple cyclic polygon of radius n if and only if either*

$$(6) \quad \sum_{i=1}^k \alpha_n(s_i) = \pi,$$

or

$$(7) \quad \alpha_n(s_j) = \sum_{i \neq j} \alpha_n(s_i)$$

for some j .

Proof. If every side uses its minor arc, the central angles $2\alpha_n(s_i)$ sum to 2π , giving (6). A simple cyclic polygon can use a major arc for at most one side. If s_j uses it, its central angle is $2\pi - 2\alpha_n(s_j)$, while all other central angles are minor; their sum is 2π exactly when (7) holds. Conversely, arranging the corresponding arcs consecutively around the circle constructs a simple cyclic polygon. $\square$

We call (6) a π -block. A positive relation with sum $\pi/2$ is called a $\pi/2$ -block. Relation (7) is a major-arc relation. A side of length $2n$ is a diameter; in this case the remaining sides form a $\pi/2$ -block.

We shall also use the strict monotonicity

$$(8) \quad \frac{d}{ds} \alpha_n(s) = \frac{1}{\sqrt{4n^2 - s^2}} > 0.$$

Thus a half-angle uniquely determines its side length at fixed radius.

Proposition 2.2 (Scaling). *For all positive integers m, n ,*

$$a(mn) \geq a(n).$$

Proof. Scale every side length of a polygon of radius n by m . All central angles are unchanged, and the new polygon has radius mn . $\square$

3. GEODETIC ANGLES AND COMPONENT CANCELLATION

For a side s , write

$$(9) \quad 4n^2 - s^2 = du^2,$$

where d is squarefree. Its complementary half-angle

$$\beta_n(s) = \frac{\pi}{2} - \alpha_n(s)$$

satisfies

$$(10) \quad \tan \beta_n(s) = \frac{u}{s} \sqrt{d}.$$

Hence $\beta_n(s)$ is a pure geodetic angle in the terminology of Conway, Radin, and Sadun [1].

We use their basis theorem and splitting theorem. In a fixed squarefree class d , write the unique basis expansion of a pure geodetic angle as

$$(11) \quad \theta = q(\theta)\pi + \sum_r c_r(\theta)\Theta_{r,d},$$

where the sum is finite and the prime-indexed basis angles $\Theta_{r,d}$ are independent modulo rational multiples of π .

Lemma 3.1 (Component cancellation). *Fix d . If*

$$\sum_i m_i \theta_i \in \pi \mathbb{Q}$$

for pure geodetic angles θ_i in class d and integers m_i , then

$$\sum_i m_i c_r(\theta_i) = 0$$

for every prime-basis element $\Theta_{r,d}$ in (11). Consequently, the signed components belonging to distinct split rational primes cancel separately.

Proof. Insert the unique expansions (11). After subtracting the rational multiple of π , uniqueness of the Conway–Radin–Sadun basis forces the coefficient of every $\Theta_{r,d}$ to vanish. $\square$

The splitting theorem further says that, in any rational relation among pure geodetic angles, the restriction to each squarefree class d is itself a rational multiple of π . These two facts will be used throughout without further comment.

The following parity principle is a useful first consequence. Besides proving the odd-sided criterion, it will later isolate the genuinely difficult part of the parity problem.

Lemma 3.2 (Non-Gaussian parity principle). *Let $\alpha_1, \dots, \alpha_m$ be half-angles of non-diameter integer chords of the same radius, none belonging to the Gaussian class $d = 1$. Let $\varepsilon_i \in \{1, -1\}$, and suppose*

$$\sum_{i=1}^m \varepsilon_i \alpha_i = tH, \quad H = \frac{\pi}{2}, \quad t \in \mathbb{Z}.$$

Then

$$\sum_{i=1}^m \varepsilon_i \equiv t \pmod{2}.$$

Proof. Group the terms by their squarefree class $d > 1$. For a side in class d , write

$$4n^2 - s^2 = du^2, \quad \xi = s + u\sqrt{-d}.$$

Then $|\xi| = 2n$ and $\arg \xi = \beta = H - \alpha$. For a fixed class put

$$m_d = \sum_{i \in d} \varepsilon_i, \quad A_d = \sum_{i \in d} \varepsilon_i \alpha_i, \quad B_d = \sum_{i \in d} \varepsilon_i \beta_i = m_d H - A_d.$$

By the splitting theorem, B_d is a rational multiple of π . On the other hand,

$$\prod_{i \in d} \xi_i^{\varepsilon_i} \in \mathbb{Q}(\sqrt{-d})^\times$$

has rational absolute value $(2n)^{m_d}$ and argument B_d . Hence its direction is a root of unity in $\mathbb{Q}(\sqrt{-d})$. For $d \neq 3$ the only roots of unity are ± 1 , so $B_d \in \pi\mathbb{Z}$ and therefore

$$\frac{A_d}{H} \equiv m_d \pmod{2}.$$

For $d = 3$, the roots of unity are the sixth roots, so initially $B_3 \in (\pi/3)\mathbb{Z}$. All classes $d \neq 3$ already have $A_d/H \in \mathbb{Z}$, and the total $\sum_d A_d/H = t$ is an integer. Thus A_3/H is an integer as well. Writing $B_3 = r\pi/3$ gives $B_3/H = 2r/3 \in \mathbb{Z}$, so $3 \mid r$ and in fact B_3/H is even. Hence $A_3/H \equiv m_3 \pmod{2}$ also in the Eisenstein class. Summing over d gives the claim. $\square$

Theorem 3.3 (Odd-sided polygons). *For a positive integer n , the following are equivalent:*

- (i) *there exists an odd-sided integer-sided cyclic polygon of radius n ;*
- (ii) *there exists such a triangle;*
- (iii) *n has a prime divisor congruent to 1 (mod 4).*

Proof. The implication (ii) $\Rightarrow$ (i) is immediate. Condition (iii) is equivalent to the existence of a nondegenerate integer right triangle with hypotenuse $2n$; inscribing it in its circumcircle gives (iii) $\Rightarrow$ (ii).

For the converse we argue contrapositively. Suppose that n has no prime divisor congruent to 1 (mod 4). Then there is no non-diameter Gaussian side: such a side would give a nontrivial identity

$$s^2 + u^2 = (2n)^2,$$

and the sum-of-two-squares criterion would force a prime $q \equiv 1 \pmod{4}$ to divide n .

Now consider an admissible polygon with k sides. In the minor-arc case, Lemma 3.2 with all signs positive and $t = 2$ gives $k \equiv 0 \pmod{2}$. For a non-diameter major-arc relation, give the target sign $+1$ and every source

side sign -1 . The signed angle sum is zero, while the signed number of sides is $1 - (k - 1) = 2 - k$; the lemma again gives k even. If the target is a diameter, the remaining $k - 1$ half-angles sum to H . Applying the lemma with $t = 1$ shows that $k - 1$ is odd, hence k is even. Thus no odd-sided polygon exists, proving (i) $\Rightarrow$ (iii). $\square$

For later normalization, put

$$(12) \quad g = \gcd(s, u), \quad A = \frac{s}{g}, \quad B = \frac{u}{g}, \quad M = \frac{2n}{g}.$$

Then

$$(13) \quad \gcd(A, B) = 1, \quad A^2 + dB^2 = M^2.$$

Only primes dividing M may occur as prime components of the angle.

4. SEMIPRIME RIGIDITY

We now specialize to $n = pq$. Instead of searching over side lengths, we first use component cancellation to reduce every possible relation to small zero-sum atoms. After the variable prime components are removed, only a short list of residual squarefree classes remains. Among these classes, only the Gaussian and Eisenstein cases have enough unit symmetry to exceed the universal hexagon bound.

4.1. Prime-component atoms. Assume first that $n = pq$ with distinct odd primes p and q .

Lemma 4.1. *If a variable prime $r \in \{p, q\}$ is inert or ramified in $K = \mathbb{Q}(\sqrt{-d})$, then $r \nmid M$ in (13).*

Proof. If r is inert and $r \mid M$, the prime ideal (r) divides $A + B\sqrt{-d}$, and hence $r \mid A, B$, contrary to coprimality. If r is ramified and $r \mid M$, then $r \mid d$. If $r \nmid A$, the valuation of $A^2 + dB^2$ at r is zero; if $r \mid A$, then $r \nmid B$ and squarefreeness of d gives valuation one. Both alternatives contradict the even valuation in M^2 . $\square$

For a split prime r , write $(r) = \mathfrak{r}\bar{\mathfrak{r}}$. If $r \mid M$, the signed exponent difference, divided by two, belongs to $\{-1, 0, 1\}$. Thus every primitive side has a normalized variable-component vector

$$(v_p, v_q) \in \{-1, 0, 1\}^2.$$

Every rational angle relation has total component vector zero by Lemma 3.1.

The remaining reduction has four steps. We first decompose arbitrary relations into minimal component atoms. We then remove the variable prime components and show that each atom has bounded residual norm. This leaves only finitely many squarefree classes d . Finally, uniform side-density bounds in the surviving classes, together with a separate treatment of major arcs, give the semiprime classification.

For orientation, the goal of the next subsections is to derive the side-density bounds needed for Theorem 1.2. Once these bounds are known, the three possible values follow from the mixing argument in Section 4.6.

Lemma 4.2 (Minimal component atoms). *Every finite zero-sum multiset of vectors in $\{-1, 0, 1\}^2$ is a disjoint union of minimal zero-sum multisets, and every minimal nonzero one has length at most 4.*

For a repeated prime $p = q$, the corresponding one-dimensional component set is contained in $\{-2, -1, 0, 1, 2\}$, and every minimal nonzero zero-sum multiset has length at most 3.

Proof. Repeatedly remove a minimal nonempty zero-sum submultiset. It remains to bound the length of one minimal atom J of nonzero vectors in

$$V = \{-1, 0, 1\}^2 \setminus \{(0, 0)\}.$$

If J contains opposite vectors v and $-v$, this pair is already a zero-sum submultiset, so minimality forces $J = \{v, -v\}$.

Assume that no opposite pair occurs. Group equal vectors and write the resulting support-minimal positive relation as

$$a_1 v_1 + \cdots + a_r v_r = 0, \quad a_i \in \mathbb{Z}_{>0},$$

with distinct directions v_i . The origin lies in the convex hull

$$0 \in \text{conv}\{v_1, \dots, v_r\}.$$

By Caratheodory's theorem in dimension two, it lies in the convex hull of at most three of these directions. Support minimality therefore gives $r \leq 3$. The cases $r = 1$ and $r = 2$ are impossible without a zero vector or an opposite pair, so $r = 3$.

For three noncollinear vectors u, v, w surrounding the origin, the primitive positive relation is obtained from

$$\det(v, w)u + \det(w, u)v + \det(u, v)w = 0$$

by choosing a common orientation, taking absolute values, and dividing by the common gcd. Every determinant has absolute value 1 or 2. A determinant has absolute value 2 only for two diagonal vectors. Two of the three determinants cannot both have absolute value 2: after fixing their common diagonal vector, the other two diagonal vectors are opposites, which has been excluded. Hence at most one primitive coefficient is 2, and the other two are at most 1. The coefficient sum, which is the length of J , is therefore at most $2 + 1 + 1 = 4$.

In one dimension, apart from opposite pairs, the only primitive minimal relations are $2 - 1 - 1 = 0$ and its negative, so the repeated-prime bound is 3. $\square$

Example 4.3 (A four-term component atom). The relation

$$2(1, 0) + (-1, 1) + (-1, -1) = 0$$

corresponds to the multiset

$$\{(1, 0), (1, 0), (-1, 1), (-1, -1)\}.$$

No proper nonempty submultiset has sum zero, so this is a minimal atom of length four. It also illustrates how the coefficient 2 can appear in the determinant argument above: one direction occurs twice, while the other two directions complete the cancellation.

We now make the residual-norm step completely explicit. Let $R = \mathbb{Z}[\sqrt{-d}]$ and $\mathcal{O}_K$ be the ring of integers of $K = \mathbb{Q}(\sqrt{-d})$.

Lemma 4.4 (Odd division in the quadratic order). *For every odd positive integer m ,*

$$(14) \quad m\mathcal{O}_K \cap R = mR.$$

Proof. If $\mathcal{O}_K = R$, there is nothing to prove. Otherwise $d \equiv 3 \pmod{4}$ and

$$\mathcal{O}_K = \mathbb{Z}\left[\frac{1 + \sqrt{-d}}{2}\right], \quad \mathcal{O}_K/R \cong \mathbb{Z}/2\mathbb{Z}.$$

Let $z \in m\mathcal{O}_K \cap R$, so $z = m\eta$ for some $\eta \in \mathcal{O}_K$. In the quotient $\mathcal{O}_K/R$,

$$0 = [z] = m[\eta].$$

Multiplication by the odd integer m is an automorphism of the group of order two. Hence $[\eta] = 0$, so $\eta \in R$ and $z \in mR$. The reverse inclusion is immediate. $\square$

Lemma 4.5 (Small residual norm). *Let J be a minimal component atom of length $m \leq 4$ for distinct odd primes p, q . Multiply the primitive side elements belonging to J and cancel the equal prime-ideal exponents above p and q . The resulting element $\xi \in R$ satisfies*

$$(15) \quad N(\xi) = 2^{2k} \quad \text{for some } 0 \leq k \leq m \leq 4.$$

In the repeated-prime case the same conclusion holds with $m \leq 3$.

Proof. For each side let $z_j = A_j + B_j\sqrt{-d} \in R$, with $N(z_j) = M_j^2$ and $M_j \mid 2pq$. Put $z = \prod_{j \in J} z_j \in R$. Component cancellation says that the valuations of the two conjugate prime ideals above p are equal, say A , and likewise those above q are equal, say B . Hence

$$(z) = (p)^A (q)^B \mathfrak{a}$$

for an integral ideal $\mathfrak{a}$, and therefore

$$z = p^A q^B \eta$$

for some $\eta \in \mathcal{O}_K$. Since $p^A q^B$ is odd and $z \in R$, Lemma 4.4 gives $\eta \in R$.

After removing the p - and q -parts, only prime factors above 2 remain. Each $M_j \mid 2pq$ contains at most one factor 2, so

$$N(\eta) = 2^{2k}$$

with $k \leq |J| = m$. Renaming η as ξ proves the claim. $\square$

If the residual element is nonreal, divide its two coefficients by their gcd. Since its norm is supported only at 2, this gives a primitive positive solution

$$(16) \quad x^2 + dy^2 = L^2, \quad L \in \{1, 2, 4, 8, 16\}.$$

The exact elementary enumeration gives, apart from the separately treated Gaussian class $d = 1$,

$$(17) \quad d \in \{3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

For a repeated prime, $L \leq 8$ and the non-Gaussian list reduces to

$$(18) \quad d \in \{3, 7, 15, 39, 55\}.$$

These finite lists are reproduced independently by exact Maple and Python computations.

4.2. The ordinary squarefree classes. Every d in (17) other than 3 satisfies $d \equiv 7 \pmod{8}$. For odd n , equation (9) then forces s and u to be even. After division by 4, the primitive norm is odd, so the individual angle has no prime-2 basis component. The same conclusion holds for any ordinary class $d \neq 1, 3$: if a nonzero prime-2 component existed, Lemma 4.5 would place d in the finite list.

The only roots of unity in $\mathbb{Q}(\sqrt{-d})$ for $d \neq 1, 3$ are ± 1 . Consequently, a rational restriction in an ordinary class is an integral multiple of

$$H = \frac{\pi}{2}.$$

A minimal atom with $k \leq 4$ sides has original-angle sum $kH - m\pi$. Positivity leaves only

$$(19) \quad k = 3, \quad \sum \alpha = H, \quad \text{or} \quad k = 4, \quad \sum \alpha = \pi.$$

Thus every ordinary class has density at most three sides per H , or one side per unit $U = \pi/6$.

For completeness we record the sharper common four-angle model used for major arcs. Up to conjugation, the nonzero exponent pairs are $(1, 0)$, $(0, 1)$, $(1, 1)$, and $(1, -1)$. Choose their two independent components so that $0 < X \leq Y < H$. The possible non-diameter half-angles are

$$(20) \quad H - X, \quad H - Y, \quad H - Y + X, \quad |H - X - Y|.$$

If $X + Y < H$, the coefficient equations show that no nonempty positive rational block exists. If $X + Y > H$, write the multiplicities in the order (20) as $m_1, \dots, m_4$. Component cancellation gives

$$m_1 = m_3 + m_4, \quad m_2 = m_4 - m_3, \quad 0 \leq m_3 \leq m_4.$$

The number of angles is $m_3 + 3m_4$, while their rational sum in units of H is $m_3 + m_4$. Hence

$$(21) \quad L_H \leq 3, \quad L_\pi \leq 6.$$

Writing an angle as $cH + aX + bY$, the functional

$$3c + 2a + 2b$$

is at least 1 on all four active vectors and at most 3 on a target vector. Therefore an ordinary major-arc relation has at most four sides.

4.3. The Gaussian and Eisenstein classes.

4.4. The Gaussian class $d = 1$. Put $H = \pi/2$. Suppose first that $p < q$ and both primes are congruent to 1 $\pmod{4}$. Let x, y be their two independent small Gaussian angles, ordered so that $0 < y < x < \pi/4$.

There are eight nonzero variable-component pairs in $\{-1, 0, 1\}^2$. Gaussian multiplication realizes these pairs, while multiplication by the four units changes only the rational H -part. Folding the resulting arguments into $(0, H]$ therefore produces exactly the eight non-diameter angles

$$x - y, \quad y, \quad H - x - y, \quad x, \quad H - x, \quad x + y, \quad H - y, \quad H - x + y.$$

Thus the list exhausts the nonzero folded component directions. Represent an angle by (c, a, b) if it equals $cH + ax + by$. Every vector in this list satisfies

$$(22) \quad 4c + 2a + b \geq 1.$$

In a rational block, the total x - and y -coefficients vanish. Therefore a sum tH containing L sides satisfies $L \leq 4t$, and in particular

$$(23) \quad L_\pi \leq 8.$$

The bound is attained because

$$(24) \quad (x - y) + y + y + (H - x - y) = H.$$

Doubling gives an octagon.

If

$$s_p^2 + u_p^2 = 4p^2, \quad s_q^2 + u_q^2 = 4q^2,$$

are oriented so that $x > y$, the corresponding side multiset at radius pq is

$$(25) \quad \left\{ \left(\frac{|s_p u_q - u_p s_q|}{2} \right)^{[2]}, (ps_q)^{[4]}, \left(\frac{u_p u_q - s_p s_q}{2} \right)^{[2]} \right\}.$$

Gaussian multiplication proves integrality. For $p = q$, the possible one-variable coefficient pairs are $(2, 0)$, $(-1, 1)$, $(1, 0)$, and $(-2, 1)$ for angles $ax + cH$. Each satisfies $a + 3c \geq 1$, so a π -block has at most six sides. If one variable prime does not split, only a subset of the one-component types occurs and the same bound holds.

4.5. The Eisenstein class $d = 3$. Put $U = \pi/6$. Suppose first that $p < q$, both primes are congruent to 1 (mod 3), and write $x = x_p$, $y = x_q$.

The Eisenstein case has the same component philosophy but a larger unit group. Multiplying the two split prime factors and the six Eisenstein units, then folding arguments into $(0, H]$, leaves finitely many coefficient triples (c, a, b) for angles $cU + ax + by$. The only change of chamber occurs when the mixed direction $x + y$ crosses U . This chamber change gives the two lists below and is precisely what changes the upper bound from 6 to 18.

If $x + y < U$, the complete non-diameter coefficient list is

$$\begin{aligned} \mathcal{E}_+ = \{ & (0, 0, 1), (0, 1, 0), (1, -1, -1), (1, -1, 1), (1, 0, 0), \\ & (1, 1, -1), (1, 1, 1), (2, -1, 0), (2, 0, -1), (2, 0, 1), \\ & (2, 1, 0), (3, -1, -1), (3, -1, 1) \}. \end{aligned}$$

The list follows by multiplying the two Eisenstein prime factors, applying the six units, and folding the arguments into $(0, H]$. Every vector satisfies

$$(26) \quad 3c + a + b \geq 1.$$

Thus a sum tU has at most $3t$ sides, and a π -block has at most 18 sides. Equality is attained by taking six copies of

$$(27) \quad x + y + (U - x - y) = U.$$

The third side length is

$$(28) \quad S = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{4}.$$

The numerator is divisible by 4 by Eisenstein multiplication, and $S > 0$ is exactly criterion (4). The resulting side multiset is

$$(29) \quad \{(qs_p)^{[6]}, (ps_q)^{[6]}, S^{[6]}\}.$$

If $x + y > U$, the complete folded list is

$$\begin{aligned} \mathcal{E}_- = \{ & (-1, 1, 1), (0, 1, 0), (0, 0, 1), (1, 1, -1), (1, 0, 0), \\ & (1, -1, 1), (2, 0, -1), (2, -1, 0), (3, -1, -1), (1, 1, 1), \\ & (2, 1, 0), (2, 0, 1), (3, 1, -1)\}. \end{aligned}$$

Every vector satisfies

$$(30) \quad c + a + b \geq 1.$$

Hence a π -block has at most six sides. If one of p, q does not split in the Eisenstein integers, the available vectors form a subset of the inactive table. For $p = q$, the one-variable specialization gives the same bounds: 18 exactly when $2x_p < U$, and otherwise 6. The boundary $2x_p = U$ is impossible because $\sin^2 x_p$ is rational, whereas $\sin^2(\pi/12) = (2 - \sqrt{3})/4$ is irrational.

Finally, a direct sine expansion gives

$$\sin(U - x_p - x_q) = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{8pq},$$

which proves the equivalence between $x_p + x_q < U$ and (4).

4.6. Mixtures and major-arc relations. The rational restrictions in an ordinary or Gaussian class are integral multiples of $H = \pi/2 = 3U$. In the Eisenstein class they are integral multiples of $U = \pi/6$.

Density argument. Side density is linear under positive mixtures of squarefree classes. This observation allows the local density estimates to be combined into a global bound: distributing the available angle among several classes cannot improve on the largest density of the participating classes. The previous subsections give the following maximal densities, measured in sides per unit U :

$$(31) \quad \rho_d \leq \begin{cases} 3, & d = 3 \text{ and } x_p + x_q < U, \\ 4/3, & d = 1 \text{ with two distinct split variable primes,} \\ 1, & \text{otherwise.} \end{cases}$$

Indeed, neutral atoms with no variable component have the ordinary bound of at most one side per unit U .

If class d contributes a positive angle A_d and k_d sides, then $k_d \leq \rho_d A_d / U$. Summing these inequalities gives

$$\sum_d k_d \leq \sum_d \rho_d \frac{A_d}{U} \leq \left(\max_d \rho_d \right) \frac{\sum_d A_d}{U}.$$

Thus a positive mixture cannot be denser than its densest participating class. Equivalently, for the purpose of the global upper bound one may spend the entire angle budget at the largest available density; mixing classes can only preserve or lower that bound. Since the class contributions are positive and a π -block has total angle $\sum_d A_d = 6U$, the global maximum is 18 in the active Eisenstein case, 8 in the remaining two-component Gaussian case,

and 6 otherwise. The constructions (24) and (27), together with the regular hexagon, show that these bounds are attained.

Lemma 4.6 (No mixed non-diameter major arc). *Every non-diameter major-arc relation is contained in one squarefree class. A diameter target reduces to a $\pi/2$ -block.*

Proof. Write a non-diameter relation as

$$\alpha_* = \sum_j \alpha_j < H,$$

and let the target angle lie in class d_0 . If a non-target class occurs, the restriction to d_0 equals the sum of the positive contributions of all non-target classes, and is therefore positive and strictly smaller than H . Every restriction is rational by the splitting theorem.

If $d_0 \neq 3$, its restriction is an integral multiple of H , impossible in $(0, H)$. If $d_0 = 3$, its restriction is U or $2U$, while every non-target ordinary or Gaussian restriction is a multiple of $H = 3U$; their sum cannot equal U or $2U$. Hence no non-target class occurs. If the target is a diameter, the remaining angles sum to H . $\square$

Inside one class, the earlier functionals give the following major-arc bounds: ordinary classes at most 4, Gaussian at most 4, active Eisenstein at most 10, and inactive Eisenstein at most 4. A diameter followed by an H -block gives at most 10 sides in the active Eisenstein case, 5 in the Gaussian case, and 4 otherwise. Every major-arc bound is therefore below the corresponding minor-arc maximum.

This completes the proof of Theorem 1.2 for odd p, q .

4.7. The even semiprime case. It remains to prove $a(2p) = 6$. Let p be odd. The same residual-norm argument leaves only

$$d \in \{1, 3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

The classes $d = 1$ and $d = 3$ are elementary. For the other ten classes, the relation pattern depends only on the ideal class of a prime ideal above p , up to conjugation.

Lemma 4.7 (Prime-replacement invariance). *Fix an imaginary quadratic field whose unit group is $\{\pm 1\}$, and fix all ideal factors supported above 2 and the ramified primes. Let P, P' be unramified split prime ideals with $[P'] = [P]$. Then the integer relation lattice among the corresponding side angles, modulo rational multiples of π , is the same for P and P' . The same holds for signed major-arc relations.*

Proof. There is $\lambda \in K^\times$ with $P' = (\lambda)P$. If an admissible side type t has ideal factor $A_t P^{e_t} \overline{P}^{f_t}$, then replacing P by P' changes its phase by

$$(e_t - f_t) \operatorname{Arg}(\lambda) \pmod{\pi}.$$

For a signed relation with coefficients m_t , the total change is

$$\operatorname{Arg}(\lambda) \sum_t m_t (e_t - f_t).$$

The sum is exactly the coefficient of the independent p -basis angle and vanishes by Lemma 3.1. The converse follows by using λ^{-1} . Conjugate ideal classes negate all variable components and preserve the relation lattice. $\square$

Reduced positive binary quadratic forms represent ideal classes; see [2, Chapters 2–3]. Therefore one represented unramified prime per reduced-form orbit under conjugation suffices. The finite ideal-class verification proceeds as follows:

- (a) enumerate all 37 reduced-form orbits in the ten classes;
- (b) choose one represented unramified prime for each orbit;
- (c) enumerate all exact solutions of $s^2 + du^2 = 16p^2$;
- (d) determine the largest π -block, $\pi/2$ -block, and major-arc relation by exact multiplication in $\mathbb{Q}(\sqrt{d}, i)$;
- (e) separately check all 19 ramified pairs $p \mid d$ and the special $d = 3$ cases.

The exact computation finds no violation of

$$(32) \quad L_\pi \leq 6, \quad L_{\pi/2} \leq 3, \quad L_{\text{major}} \leq 4.$$

The regular hexagon gives equality. The remaining case $p = 2$, namely $n = 4$, is checked directly. Thus $a(2p) = 6$ for every prime p .

5. GROWTH ALONG FIXED POWERS

The semiprime theorem might suggest that few prime divisors force bounded behavior. The next construction shows that exponents alone can have the opposite effect.

Fix $N \geq 2$ and write

$$(33) \quad 4N^2 - 1 = dv^2,$$

where d is squarefree and $v \geq 1$. Put

$$z = 1 + v\sqrt{-d}, \quad N(z) = 4N^2,$$

and write

$$z^j = x_j + y_j\sqrt{-d} \quad (j \geq 1).$$

Lemma 5.1 (The quadratic chord tower). *For every $j \geq 1$, both x_j and y_j are divisible by 2^{j-1} . If*

$$X_j = \frac{x_j}{2^{j-1}}, \quad Y_j = \frac{y_j}{2^{j-1}},$$

then

$$(34) \quad X_j^2 + dY_j^2 = 4N^{2j}.$$

In particular, $|X_j|$ is an integer chord length at radius N^j . Moreover, if

$$\theta = \arg z \in (0, \pi/2), \quad \tau = \frac{2\theta}{\pi},$$

then τ is irrational and the associated half-angle is

$$(35) \quad \alpha_j := \alpha_{N^j}(|X_j|) = \arcsin |\cos(j\theta)|.$$

Proof. The real parts satisfy the trace recursion

$$x_{j+2} = 2x_{j+1} - 4N^2x_j, \quad x_0 = x_1 = 1.$$

With $X_j = x_j/2^{j-1}$ this becomes

$$(36) \quad X_{j+2} = X_{j+1} - N^2X_j, \quad X_1 = 1, \quad X_2 = 1 - 2N^2,$$

so all X_j are integers. The imaginary parts satisfy the same trace recursion, with $y_1 = v$ and $y_2 = 2v$; after division by 2^{j-1} the same integral recurrence applies. Hence $2^{j-1} \mid y_j$. Dividing $N(z^j) = (4N^2)^j$ by 4^{j-1} proves (34).

Since $z/(2N) = e^{i\theta}$,

$$\frac{X_j}{2N^j} = \cos(j\theta),$$

which gives (35). Finally, if θ/π were rational, then $e^{i\theta}$ would be a root of unity and

$$2 \cos \theta = \frac{1}{N}$$

would be a rational algebraic integer. It would therefore be an integer, which is impossible for $N \geq 2$. Hence τ is irrational. $\square$

For $e \geq j$, scaling the chord in Lemma 5.1 gives the integer side

$$(37) \quad s_{e,j} = N^{e-j} |X_j|$$

at radius N^e , with the same half-angle α_j .

We use one elementary fact about the Stern–Brocot path of an irrational number in $(0, 1)$.

Lemma 5.2 (Odd–odd bracketing edges). *Let $\tau \in (0, 1)$ be irrational. Along its Stern–Brocot path there are infinitely many bracketing edges*

$$(38) \quad \frac{a}{b} < \tau < \frac{c}{d}, \quad bc - ad = 1,$$

for which both numerators a and c are odd. The denominators of these edges are unbounded.

Proof. At every Stern–Brocot step the two endpoint numerators cannot both be even, because the determinant is 1. If they have different parity, their median has odd numerator. Replacing the even endpoint therefore produces an odd–odd edge. If this never happened again, the algorithm would have to replace the odd endpoint at every subsequent step while keeping the even endpoint fixed. The successive medians would then converge to that fixed rational endpoint, forcing τ to equal it. This contradicts irrationality. After an odd–odd edge the median has even numerator, so the same argument restarts. Thus odd–odd edges occur infinitely often. Their denominators are unbounded because the Stern–Brocot intervals containing an irrational point form a strictly nested infinite sequence. $\square$

Proof of Theorem 1.3. Choose an odd–odd edge (38). Because the endpoints are Farey neighbors,

$$0 < b\tau - a < \frac{1}{d} < 1, \quad 0 < c - d\tau < \frac{1}{b} < 1.$$

Since a and c are odd, these are precisely the distances of $b\tau$ and $d\tau$ from the nearest odd integers. Formula (35) therefore gives, with $H = \pi/2$,

$$\alpha_b = H(b\tau - a), \quad \alpha_d = H(c - d\tau).$$

The determinant identity now yields

$$(39) \quad d\alpha_b + b\alpha_d = H(bc - ad) = H.$$

Doubling (39) gives a π -block with $2(b + d)$ sides. By (37), both required chord lengths are integers at the common radius N^e , where $e = \max\{b, d\}$. Hence

$$(40) \quad a(N^e) \geq 2(b + d) \geq 2e.$$

The odd-odd edges have unbounded denominators, so (5) follows. By Proposition 2.2, the sequence $a(N^e)$ is nondecreasing in e . Since the construction above makes it unbounded along an infinite subsequence, monotonicity implies that for every M there is an e_0 such that $a(N^e) \geq M$ for all $e \geq e_0$. Hence $a(N^e) \rightarrow \infty$. $\square$

Example 5.3 (A thirty-eight-gon at radius 8192). Take $N = 2$. Then $d = 15$, $v = 1$, and $z = 1 + \sqrt{-15}$. For

$$\tau = \frac{2 \arg z}{\pi}$$

one has the odd-odd Farey edge

$$\frac{5}{6} < \tau < \frac{11}{13}, \quad 6 \cdot 11 - 5 \cdot 13 = 1.$$

The scaled power coefficients are

$$\frac{\Re(z^6)}{2^5} = -7, \quad \frac{\Re(z^{13})}{2^{12}} = -2339.$$

At the common radius $2^{13} = 8192$, the first chord scales by 2^7 and has length 896, while the second has length 2339. Thus

$$26\alpha_{8192}(896) + 12\alpha_{8192}(2339) = \pi,$$

so

$$\{896^{[26]}, 2339^{[12]}\}$$

is a thirty-eight-gon and $a(8192) \geq 38$.

6. PARITY WITH ONE GAUSSIAN SPLIT PRIME

The growth theorem is still compatible with a parity restriction. We first separate the global geometric issue from the one-dimensional Gaussian optimization.

Put $H = \pi/2$. A *Gaussian H -block* is a multiset of non-diameter sides from the squarefree class $d = 1$ whose half-angles sum to H ; a *Gaussian π -block* is defined similarly. Let $H_G(n)$ and $P_G(n)$ be their maximum cardinalities, with value 0 if the corresponding class is absent.

Lemma 6.1 (Reduction of an odd maximum). *If $a(n)$ is odd, then an extremal polygon is a pure Gaussian minor-arc polygon. In particular, its side count is at most $P_G(n)$.*

Proof. By Theorem 3.3, an odd polygon must involve the Gaussian class. Consider first a minor-arc π -block. The splitting theorem makes the Gaussian subtotal a multiple of H . If other squarefree classes occur, positivity forces the Gaussian and non-Gaussian subtotals both to equal H . Let their side counts be g and r . Doubling either H -block gives an even minor-arc polygon with $2g$ or $2r$ sides. Since $g + r$ is odd, $g \neq r$, and

$$\max\{2g, 2r\} > g + r.$$

Thus a mixed odd minor-arc polygon cannot be extremal. An odd extremal minor-arc polygon must therefore be purely Gaussian.

If a diameter occurs, the remaining sides form an H -block with, say, $r \geq 2$ sides. Doubling that block gives a $2r$ -gon, which is larger than the original $(r + 1)$ -gon.

Finally consider a non-diameter major-arc relation

$$\alpha_* = \sum_j \alpha_j < H.$$

If the target is not Gaussian, the Gaussian restriction is a multiple of H lying in $(-H, 0]$, hence it is zero; no Gaussian source side occurs, and the parity statement for the classes $d > 1$ makes the total number of sides even. If the target is Gaussian, its Gaussian restriction lies in $[0, H)$ and must again be zero. Thus every source side is Gaussian and the relation is pure Gaussian. Writing $4n^2 - s_*^2 = u_*^2$, the complementary chord u_* is an integer and has half-angle $H - \alpha_*$. The source sides together with this complementary chord form an H -block with as many sides as the original major-arc polygon; doubling it gives a strictly larger even minor-arc polygon. Hence an odd maximum can only be a pure Gaussian π -block. $\square$

Assume now that n has exactly one distinct prime divisor $q \equiv 1 \pmod{4}$, and write $q^e \parallel n$. Choose a Gaussian prime

$$\varpi = A + iB, \quad q = \varpi \overline{\varpi}, \quad 0 < B < A,$$

and put

$$(41) \quad \theta = \arg \varpi, \quad \tau = \frac{4\theta}{\pi} \in (0, 1).$$

The number τ is irrational. Indeed, a rational value would make $\varpi/\overline{\varpi} = e^{2i\theta}$ a root of unity in $\mathbb{Q}(i)$; since $0 < \theta < \pi/4$, this is impossible.

For $1 \leq j \leq e$, let

$$\ell_j = \lfloor j\tau \rfloor, \quad r_j = j\tau - \ell_j.$$

Lemma 6.2 (The rank-one Gaussian angle set). *The non-diameter Gaussian half-angles at radius n are exactly*

$$(42) \quad Hr_j, \quad H(1 - r_j), \quad 1 \leq j \leq e.$$

Proof. Let $s + iu$ be a Gaussian integer with $s^2 + u^2 = (2n)^2$. A prime $r \equiv 3 \pmod{4}$ is inert in $\mathbb{Z}[i]$, so its exponent in $s + iu$ is forced and contributes no argument. The prime 2 is ramified and contributes only a multiple of H , which can be absorbed into the unit. For the unique split prime $q = \varpi \overline{\varpi}$,

write the exponents of ϖ and $\overline{\varpi}$ in $s + iu$ as a and $2e - a$. Their contribution to the argument is

$$(a - (2e - a))\theta = 2(a - e)\theta.$$

Thus, modulo H , every nontrivial Gaussian direction is represented by $2j\theta = Hj\tau$ with $1 \leq j \leq e$, up to sign. Folding into $(0, H)$ gives Hr_j and $H(1 - r_j)$. Conversely, choosing exponents $e + j$ and $e - j$ (and an appropriate Gaussian unit) realizes both folded directions. Since τ is irrational, none of these angles is 0 or H . $\square$

The primes 2 and those congruent to 3 (mod 4) therefore change only the rational scale of the corresponding chords; they create no new Gaussian direction.

Let

$$(43) \quad \frac{a}{b} < \tau < \frac{c}{d}$$

be the two neighbors of τ in the Farey sequence of order e . Then

$$bc - ad = 1, \quad b, d \leq e, \quad b + d > e.$$

Put

$$h = b + d, \quad C = a + c.$$

Lemma 6.3 (Exact Gaussian block maxima). *With the notation above,*

$$(44) \quad H_G(n) = h, \quad P_G(n) = 2h.$$

Proof. Represent the two angle types in (42) by

$$v_j = (-\ell_j, j), \quad v'_j = (\ell_j + 1, -j),$$

so that a vector (x, y) represents the angle $H(x + y\tau)$. An H -block therefore sums to $(1, 0)$ and a π -block to $(2, 0)$.

The mediant C/h lies strictly between the Farey neighbors (43). For every $1 \leq j \leq e$, the function $x \mapsto \lfloor jx \rfloor$ is constant on this Farey interval: a change would occur at a rational with denominator at most $j \leq e$, contradicting that the two endpoints are adjacent in the Farey sequence of order e . Consequently

$$\ell_j = \left\lfloor \frac{Cj}{h} \right\rfloor,$$

and hence

$$(45) \quad 1 \leq Cj - h\ell_j \leq h - 1.$$

Define the integral linear functional

$$F(x, y) = hx + Cy.$$

By (45), every available vector v_j and v'_j has $1 \leq F \leq h - 1$. Thus an H -block has at most h sides and a π -block at most $2h$ sides.

For the lower bound, Farey adjacency gives

$$d(-a, b) + b(c, -d) = (1, 0).$$

Here $(-a, b) = v_b$ and $(c, -d) = v'_d$, because $a/b < \tau < c/d$. Hence d copies of the first angle and b copies of the second form an H -block with $b + d = h$ sides. Doubling it gives a π -block with $2h$ sides. This proves (44). $\square$

Proof of Theorem 1.4. If $\omega_1(n) = 0$, Theorem 3.3 shows that every admissible polygon already has even side count. Suppose $\omega_1(n) = 1$. If $a(n)$ were odd, Lemma 6.1 would reduce an extremal polygon to a pure Gaussian π -block. But Lemma 6.3 shows that the largest such block has the even cardinality $P_G(n) = 2h$. An odd block cannot attain this maximum, a contradiction. Therefore $a(n)$ is even. $\square$

Remark 6.4. The proof identifies the precise point at which the one-prime hypothesis is used. With several distinct primes $q \equiv 1 \pmod{4}$, Gaussian angles are indexed by a multidimensional exponent box rather than an interval. The Farey dual functional is then replaced by a bounded zero-sum problem in a higher-dimensional lattice. The general conjecture that $a(n)$ is always even remains open.

7. COMPUTER-ASSISTED VERIFICATION AND REPRODUCIBILITY

The semiprime ancillary directory contains exact Maple programs for the odd structural verification and the even ideal-class verification, an independent Maple cross-check comparing the recursive maxima with a separate finite Cartesian enumeration, and exact Python reimplementations of the odd and even computations. All these calculations use exact integer arithmetic and exact multiplication in the basis

$$1, \quad \sqrt{d}, \quad i, \quad i\sqrt{d}.$$

No floating-point trigonometric comparison is used.

For a fixed ordered list $V = (v_1, \dots, v_h)$ of distinct phase vectors, the block recursion processes the types in order and tries multiplicities $0, 1, 2, \dots$ for each type. It prunes only after the exact accumulated phase has crossed the target π , $\pi/2$, or the distinguished major-arc target. Since every remaining angle is positive, no extension of a pruned node can return to equality. Consequently every admissible multiplicity vector is visited exactly once. The separate cross-check independently derives a finite upper bound for each multiplicity and exhausts the resulting Cartesian box; it agrees with the recursive maximum in every case. The monotonicity (8) ensures that a target half-angle corresponds to a unique side length.

The verification has three independent layers.

- (i) The primary Maple certificates verify the odd structural reduction and the even ideal-class computation. They recover the 11 candidate squarefree classes, all required linear-functional bounds, the 37 unramified reduced-form orbits and the 19 ramified pairs, with no violation of (32).
- (ii) An independent Maple cross-check replaces the recursive search by a finite Cartesian enumeration with separately derived multiplicity bounds. The two methods give the same maximum in every case.
- (iii) Independent Python implementations reproduce the odd and even candidate classes, orbit counts, and extremal bounds using exact integer arithmetic and a separately written code path.

The verification is therefore supported by independent implementations and does not rely on numerical trigonometric comparisons. Computation enters only after the theoretical semiprime reduction has reduced the remaining

questions to finite exact searches. No computation is used in the proofs of Theorems 1.3 and 1.4.

A separate end-to-end numerical validation, not used as a logical step in the proof, gave

$$a(411) = 6, \quad a(422) = 6, \quad a(445) = 8, \quad a(469) = 18, \quad a(529) = 6.$$

The exact Maple and Python programs, the independent cross-check, recorded outputs, and reproduction instructions are supplied as Online Resource 1. They are intended to accompany the public preprint as ancillary files as well, so that the finite verification can be reproduced without relying on unpublished source code.

Online Resource 1. Exact verification package for the semiprime classification, containing the Maple certificates, independent Python reimplementations, an independent Maple cross-check, recorded outputs, reproduction instructions, and a SHA-256 manifest.

8. EXAMPLES AND CONSEQUENCES

We record a few examples illustrating the main results.

Example 8.1 (Eisenstein jump). For $p = 7$ and $q = 13$,

$$(s_7, u_7) = (2, 8), \quad (s_{13}, u_{13}) = (1, 15).$$

Formula (28) gives $S = 61$. Thus

$$\{7^{[6]}, 26^{[6]}, 61^{[6]}\}$$

is the side multiset of an eighteen-gon of radius 91, and $a(91) = 18$.

Example 8.2 (Gaussian jump). For $p = 5$ and $q = 13$, the Gaussian construction gives

$$\{32^{[2]}, 50^{[4]}, 66^{[2]}\},$$

so $a(65) = 8$.

Corollary 8.3. *For a prime p ,*

$$a(p^2) = 18$$

exactly when $p \equiv 1 \pmod{3}$ and $2x_p < \pi/6$; otherwise $a(p^2) = 6$. In particular,

$$a(49) = 18, \quad a(169) = 18, \quad a(361) = 6.$$

On the other hand, Theorem 1.3 gives $a(p^e) \rightarrow \infty$ as $e \rightarrow \infty$, while Theorem 1.4 shows that every one of these maxima is even.

Together, Theorems 1.2 and 1.3 show the different behaviour of semiprime radii and high powers: the semiprime case has a finite exact classification, whereas no fixed prime support bounds the number of sides when exponents grow.

9. OUTLOOK

Two questions remain. First, the parity theorem suggests the conjecture

$$a(n) \equiv 0 \pmod{2} \quad (n \geq 1).$$

The one-split-prime case is exactly one-dimensional and the Farey argument solves it. Several split Gaussian primes lead instead to a higher-dimensional bounded zero-sum problem; resolving that problem appears to be the main obstruction to a general parity theorem.

Second, Theorem 1.3 gives the universal lower bound $\limsup a(N^e)/e \geq 2$, but it does not determine the true growth rate for any fixed N . Even the binary sequence $a(2^e)$ presents a concrete next problem: determine its order of growth, or sharpen the constant 2 in the limsup bound.

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and independently checked every proof, calculation, and reference. No AI-generated output is used as mathematical evidence, and no research data or figures were generated by AI. The author reviewed the final manuscript and takes full responsibility for its content.

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